Makers

“Component Sorter” project

Final report

Project context, stakeholders and stakes

In laboratories and practical-work rooms, electronic components are used every day for all kinds of experiments. While building circuits is often a stimulating activity, dismantling them and putting the components away is a much more tedious task.

After each session, the components have to be sorted, and then their storage location has to be found. This process can be long, repetitive and error-prone, which sometimes leads to components that are still usable being thrown away.

To address this, we propose an innovative automated storage system dedicated to electronic components. In a first stage, our solution focuses exclusively on integrated circuits, in order to limit the scope of the project and optimise the performance of the system.

Several stakeholders are involved in this project:

- the students and teachers who use the practical-work rooms,
- the laboratory managers,
- and, more broadly, any institution that wants to improve the management of its electronics stock.

The stakes are multiple:

- saving time during the dismantling and storage phases,
- reducing classification and storage errors,
- organising the components for faster and more structured access,
- and finally, encouraging the reuse of equipment, with sustainability in mind.

Functional specifications

Phase A: Installing the system

- **FA1:** The system must be easy to install.

Phase B: Selecting and identifying an integrated circuit

- **FB1:** The system must be able to pick a component from a tray of unsorted components.
- **FB2:** The system must precisely identify the selected integrated circuit.
- **FB3:** If the selected component is unknown, the system must assign it a new storage compartment if one is still available, and otherwise put it back in the tray. If it is known, the system must store it

in the right compartment.

Phase C: Storing in a compartment

- **FC1:** The system must carry the component to the corresponding compartment and drop it safely into that compartment.
- **FC2:** The system must automatically return to its initial position (that is, above the tray of unsorted components).

Other functions D

- **FD1:** The system must be able to sort as many components as possible.
- **FD2:** The system must be intuitive and simple to use.

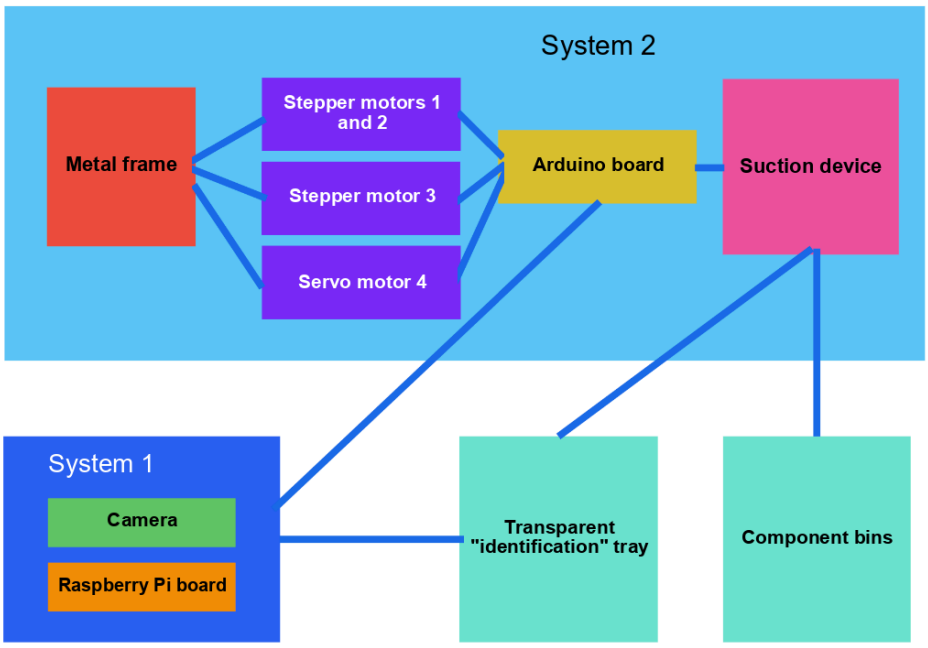
Physical and technological principles: the key points in this field

Automated motion: motors move a suction device along the X, Y and Z axes, following the instructions given by the code. This relies on principles of mechanics and automatic control.

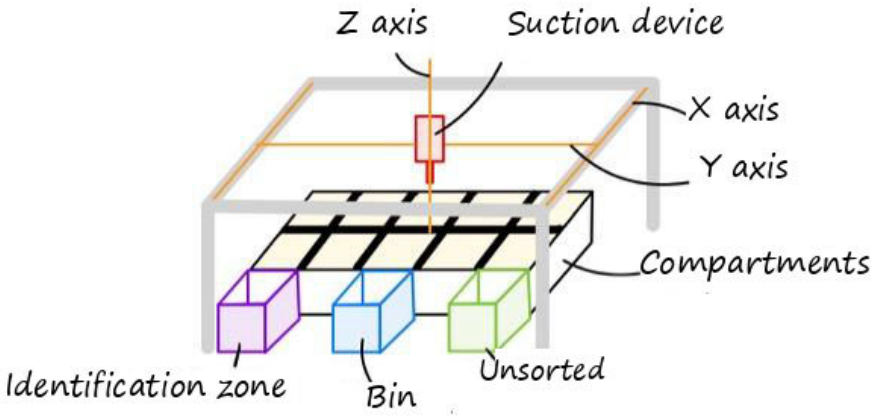
Analysis and classification of the components: the device uses machine-learning algorithms to analyse the text printed on the components and determine where to store them. This process relies on advanced natural-language processing and computer-vision techniques.

System design file: mechanics, electronics, software

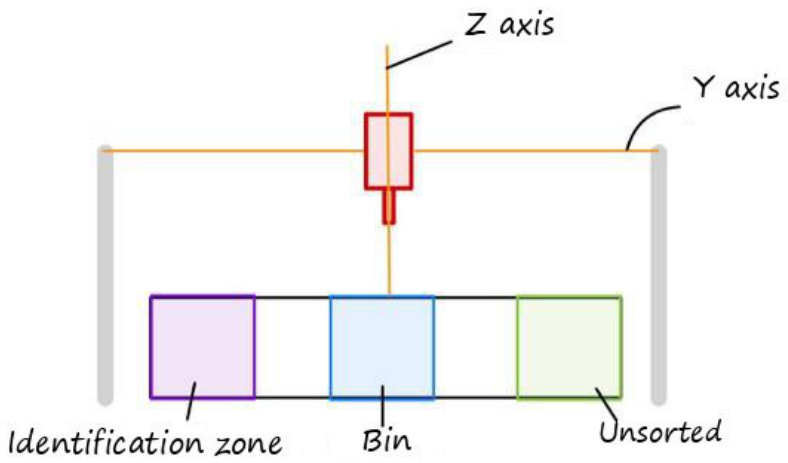
System architecture



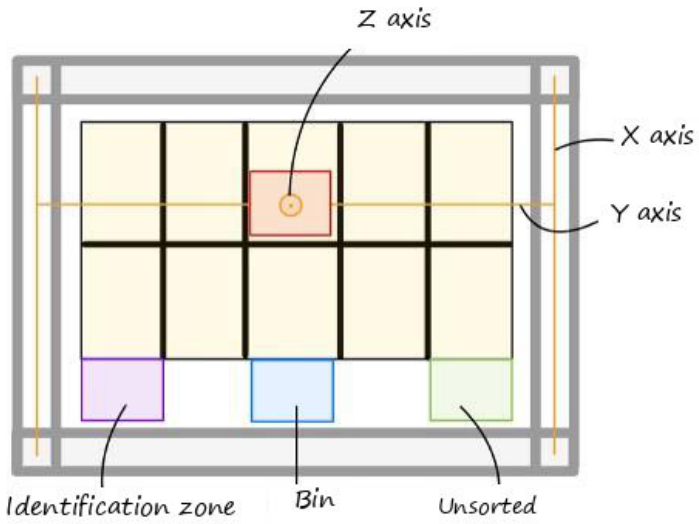
Plans and drawings of the system



System, 3D view



Side view



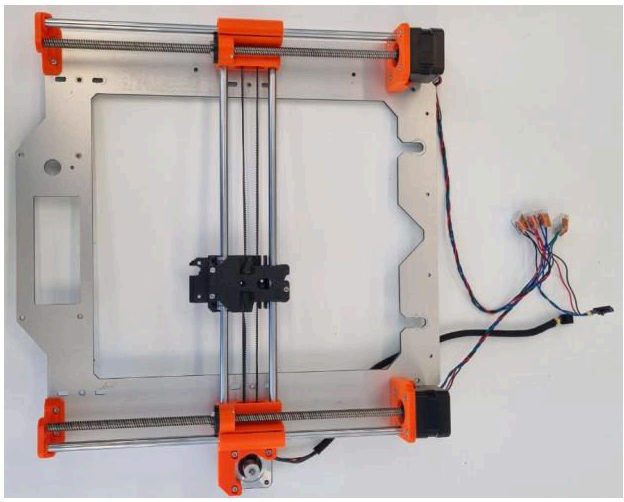
Top view

Pre-sizing calculations

We chose to work with an Arduino UNO R3 board coupled (for the X and Y motors) with TMC2209 drivers, to meet the power requirements of our architecture (an unsuccessful attempt had been made with DRV8825 drivers), but also because these two tools are the simplest and best suited for systems using stepper motors and servo motors.

Since our system relies in part on an AI-based image-recognition algorithm and therefore needs significant computing power, a Raspberry Pi board seemed the most suitable. As these algorithms are demanding in computing capacity, we chose the Raspberry Pi 5, the latest generation, suited to running AI algorithms. The Camera Module V3 is the camera associated with this type of board.

The XY rail had to be several tens of centimetres long, so that components could be stored in multiple bins of a few centimetres each. The XY frame of the MK3S printer turned out to be ideal, as it is robust and of a satisfactory size (30 cm × 30 cm).



In addition, we estimated that our carriage should be able to travel along the diagonal of the XY plane in a reasonable time of 15 seconds. For efficiency and reliability reasons, we adapted the rest of the XZ axes of an MK3S printer to our project. The motors we obtained do allow such a movement along the diagonal.

For the vacuum suction, we needed something able to lift small electrical components: resistors, integrated circuits, etc. A consumer vacuum pen turned out to be more than sufficient, given the low power required.

The movement of the pen along the Z axis had to happen in a reasonable time, which we estimated at 10 seconds. The Z-axis motor and rail, recovered from a previous student project, with a torque of 25 kg, proved very effective.

Software components

There are software components on the Raspberry Pi board and on the Arduino board.

Raspberry Pi

- `Main.py`: the main program, run when the system starts. It sends the orders to the Arduino board.
- `Prise_photo.py` (take photo): connects to the camera and takes a photo while an LED is on (acting as a flash).
- `Otstu_threshold.py`: crops the image so that the component is centred.
- `K_voisin.py`: uses the image returned by `Otstu_threshold.py` to apply a filter to the cropped image.
- `detect.py`: uses the cropped and filtered image returned by `K_voisin.py` to read the text on the image.

Arduino

- `prototype.ino`: the main program, containing the control algorithm for all the motors, in response to the commands sent by the Raspberry Pi.

Tests and lessons learned on the prototype

Testing the text-recognition code

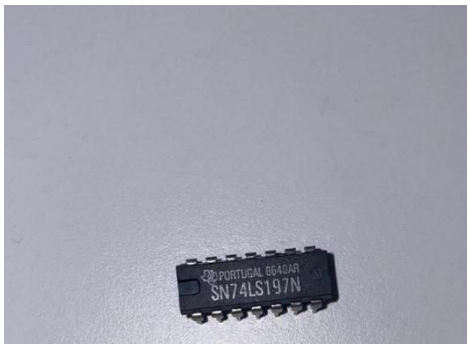


Photo of the component



Image of the component after cropping



Image of the component after cropping and filtering

```
Image enregistrée sous : otsu_result2.jpg
Cropped :  150  | Otsu :  150  | texte :  N7ALS1 g7N

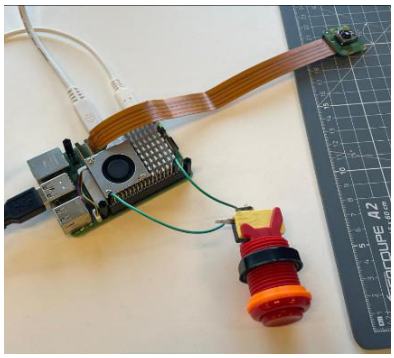
[('76', 'N7ALS1'), ('45', 'g7N')]
Image enregistrée sous : otsu_result2.jpg
Cropped :  150  | Otsu :  152  | texte :  ~ agomg pani
N74LS19 7N

[('0', '~'), ('0', 'agomg'), ('2', 'pani'), ('68', 'N74LS19'), ('81', '7N')]
~ agomg pani
N74LS19 7N
```

Result of the text reading (program output; “Image enregistrée sous” = “Image saved as”)

Our code successfully identified “N74LS197N”, with a confidence of 68% for “N74LS19” and 81% for “7N”. This result is very promising: for the first prototype, we plan to sort the following 3 components: SN74LS197N, SN74LS22N and SN74LS245N. The confidence level is therefore sufficient to identify and tell these 3 components apart.

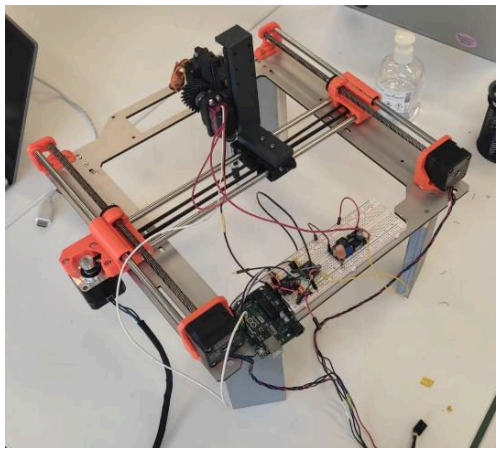
Testing the camera



The camera connected to the Raspberry Pi

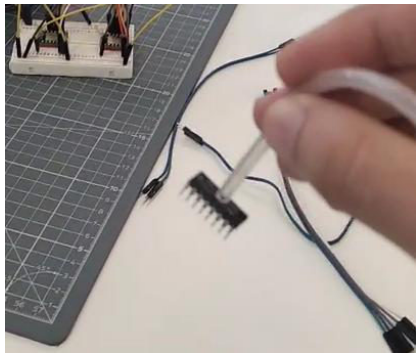
The camera was tested successfully. It is properly integrated into the system and optimally positioned, ensuring high-quality image acquisition, which is essential for the visual processing and analysis planned in the project.

Testing the X, Y and Z axis motor code



The X, Y and Z axis motors were tested and work perfectly. We manage to move the suction device to the desired positions in the X-Y plane, and the vertical movement is also a success.

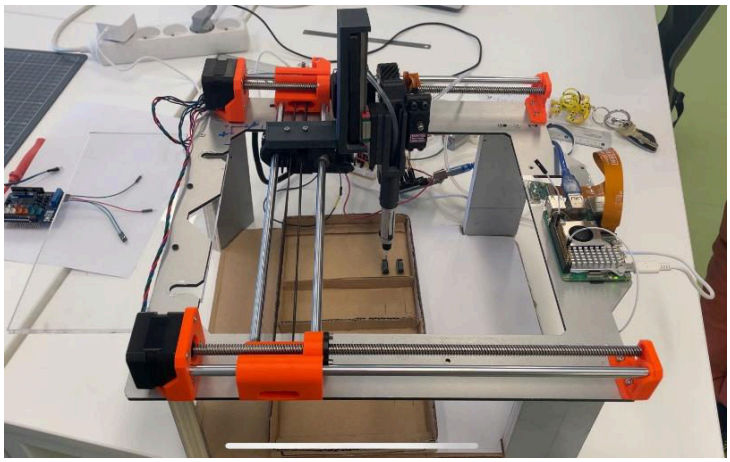
Testing the suction device



The suction device was tested successfully. It sticks properly to the intended surfaces and fully fulfils its role within the motion system.

Testing the whole system

We had to modify our electronic circuit (which links the software and mechanical parts) to make it more robust to the high power load it had to withstand. This led us to rethink the layout of the components and the wiring. More precisely, we redesigned the way the components were interconnected to optimise heat dissipation, limit losses and ensure better reliability of the circuit under heavy load.



Possible improvements and perspectives for continuing the project

We have identified a number of improvements to make to our system:

- Add other types of identifiable components (e.g. resistors, capacitors)
- Increase the sorting speed by increasing the power of the motors
- Automate the picking of the components, for example by adding a conveyor belt that would bring them one by one
- Add a display panel (e.g. an LED screen) to give the user a visual aid, so that they can see which component each compartment corresponds to once sorting is finished
- Add a ventilation system to prevent the boards from overheating
- Increase the number of sorting compartments (to 16, or even 32)
- Add mounting feet and a compartment-recognition system, so that the device can be integrated directly into the drawers of the FabLab

List of purchased components and their suppliers

With both an economical and an eco-responsible approach in mind, our goal is to reuse as many used components available at the CentraleSupélec FabLab as possible. This approach aims to limit material costs while making the most of existing resources. At this stage of the project, only the vacuum motor has been ordered.

The list of components and their suppliers is given in the bill of materials below (supplier links are in the team's original shared spreadsheet, available on request).

Bill of materials

#	Item	Reference	Supplier	Qty	Unit cost	Total cost
1	Z axis, bottom right (MK3)		Prusa Research	1	€3.85	€3.85
2	Z axis, bottom left (MK3)		Prusa Research	1	€3.85	€3.85
3	Z axis, top right (MK3)		Prusa Research	1	€3.85	€3.85
4	Z axis, top left (MK3)		Prusa Research	1	€3.85	€3.85
5	Z screw cover (MK2S)		Prusa Research	2	€3.85	€7.70
6	X carriage D2 set (MK3S)		Prusa Research	1	€7.75	€7.75
7	Linear bearing LM8UU		Prusa Research	6	€2.09	€12.54
8	Pulley T16-2GT		Prusa Research	1	€3.08	€3.08
9	Printer frame		Prusa Research	1	€84.90	€84.90
10	Smooth rod, Z axis		Prusa Research	2	€11.64	€23.28
11	Smooth rod, X axis		Prusa Research	2	€11.99	€23.98
12	Stepper motor, Z axis right		Prusa Research	1	€38.89	€38.89
13	Stepper motor, Z axis left		Prusa Research	1	€38.89	€38.89
14	Stepper motor, extruder axis		Prusa Research	1	€38.89	€38.89
15	X motor mount (MK3)		Prusa Research	1	€3.85	€3.85
16	Z-axis bracket and end stop		CentraleSupélec (Onshape design)	1	€0	€0
17	Vacuum suction pen		Amazon	1	€28.16	€28.16
18	Arduino UNO R3		Arduino Store	1	€28.80	€28.80
19	Stepper motor driver	DRV8825	Gotronic	2	€15.50	€31.00
20	Electrolytic capacitor	10YXJ100MTA5X11	Farnell	2	€0.10	€0.21
21	Raspberry Pi 5	SC1112	Amazon	1	€97.38	€97.38
22	Raspberry Pi active cooler	SC1148	Amazon	1	€10.73	€10.73
23	Raspberry Pi power supply	SC1152	Amazon	1	€20.89	€20.89
24	Raspberry Pi Camera Module V3	PICAM3X	Kubii	1	€30	€30
25	Servo motor	DS3225	Abra Electronics	1	€29.80	€29.80
26	12 V power adapter		Amazon	1	€16.99	€16.99
27	Adjustable potentiometer	3590S	Amazon	1	€13.99	€13.99